Roll No. ....

TPE–404

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

APPLIED SEDIMENTOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1\. (a) (i) Water flowing through a pipe has a dynamic viscosity of 8.9×10^{-4} Pa.s and density of 1000 kg/m^3. The pipe has an internal diameter of .025 m and the water flows at an average speed of .15 m/s. What is the Reynolds' number for the flow ?

(ii) Explain Diagenesis. (CO1)

OR

(b) What are the differences between physical weathering and chemical weathering ? What are the main agents responsible for physical weathering and chemical weathering ? (CO1)

P. T. O.

2. (a) (i) A syrup flows at a rate of 4.0 m^3s^{-1} in a circular pipe of 6 cm, syrup has a density of 1300 kgm^{-3} and viscosity of 17 Pass. Find whether the flow is laminar or turbulent.

(ii) Describe mode of sediment transport. (CO1, CO2)

OR

(b) Explain Sediments packing, Permeability and types of permeability. (CO1, CO2)

3. (a) (i) Describe Reynolds' number and with the help of neat diagram, explain Laminar and Turbulent flow.

(ii) If a fluid with a viscosity of 0.4 Ns/m^2 and relative density of 900 kg/m^3 flows through a pipe of 20 mm with a velocity of 2.5 m, calculate the Reynolds' number. (CO1, CO2)

OR

(b) With the help of table explain, Wentworth grain size classification (CO1, CO2)

4. (a) Explain what clastic sedimentary rock is. Write the name of two clastic sedimentary rocks. Give the classification of clastic sedimentary rocks. (CO1, CO2)

OR

(b) What is Physical Weathering ? Describe three processes that contribute to the weathering of rocks. (CO1, CO2)

5. (a) Explain what do you understand by kurtosis and what does it represents ? Draw the kurtosis curve for (i) sorted (ii) poor sorted (iii) well sorted. (CO1, CO2)

OR

(b) Explain porosity and types of porosity. Discuss all the parameters on which porosity of rocks depend. (CO1, CO2)

Roll No. ....

TPE–413

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MUD LOGGING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the three different sectors of Oil and Gas Industry' ? Explain each in detail. (CO1)

OR

(b) What are the various stages of up-stream oil gas life cycles ? Explain each in detail. (CO1)

2. (a) What is mud logging ? Why is it required ? How is it different from Well logging ? What are its limitations ? (CO1)

OR

(b) Explain mud logging principle. Describe different components of Mud Logging units. What are the various sensors available in recording the data ? (CO1)

P. T. O.

3. (a) Explain rig up and rig down in mud logging. What are the precautions to be taken during this process ? (CO1)

OR

(b) How is the mud logging used for Formation Evaluation ? Explain Lag up, lag-down, and complete cycle in detail. (CO1)

4. (a) Explain Lag time and Lag strokes. How is it calculated ? (CO2)

OR

(b) What are the various mud logging parameters recorded during data acquisition ? What are the various sensors used for recording these parameters ? (CO2)

5. (a) What is mud log presentation ? What are the various components of mud log to be put in various tracks for interpretation ? (CO2)

OR

(b) What are the uses of drilling cutting analysis during mud logging ? What is shale shaker and why is it required ? (CO2)

Roll No. ....

TMA–403

B. TECH. (PETROLEUM ENGG.)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

NUMERICAL AND GEOSTATISTICAL METHODS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the following terms : (CO1)

Mean, median, mode, range, variance and standard deviation.

OR

(b) The following are the values of two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at

P. T. O.

various time period (x) on individuals. Find the regression line of y on x and on y also find the coefficient of correlation : (CO1)

Time Period (x)	Diastolic blood pressure (y)
0	72
5	67
10	70
15	65
20	66

2. (a) The number of airplane landings at an airport in a minute (X) and their probability are given by : (CO1)

X_i	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the mean and standard deviation.

OR

(b) A multiple choice test consists of 8 questions with 3 options of each question in which only one is correct. A student answers each question by rolling a balanced die and checking the first answer if he gets 1 or 2, the second answer if he gets 3 or 4 and the third answer if he gets 5 or 6. To get a distinction, the student must secure at least 75% correct answer. If there is no negative marking, what is the probability that the student secures a distinction. (CO1)

3. (a) The first moments about the working mean 28.5 of a distribution are 0.294, 7.144, 42.409, and 454.98. Calculate the moments about the mean, also calculate β_1, β_2, and comment upon the skewness and kurtosis of the distribution. (CO2, CO1)

OR

(b) A sample of 100 battery cells is tested to find length of life with mean 10 hours and standard deviation 4 hours. Assuming that the data is normally distributed, what percentage of battery cells are expected to have life : (CO2, CO1)

(i) more than 13 hours

(ii) less than 5 hours.

4. (a) Using least square method to fit a curve $y = ax^2 + b$ for the following data : (CO2)

x	y
1	10
3	5
5	20
10	25
15	22

OR

(b) Two samples are drawn from two normal populations. From the data given below. Test whether the two samples have the same variance at 5% level of significance : (CO2)

Sample I	Sample II
60	61
65	66
71	67
74	85
76	78
82	63
85	85
87	86
–	88
–	91

Tabulated value at 5% level of significance and 9 degree of freedom is 3.68.

5. (a) Following are the data on the drying time of a certain varnish and the amount of an additive that is intended to reduce the drying time :

(CO1, CO2)

Amount of varnish additive groups (x)	Drying time (y in hours)
0	12.0
1	10.5
2	10.0
3	8.0
4	7.0
5	8.0
6	7.5
7	8.5
8	9.5

(i) Fit a simple linear regression line.

(ii) Use the result of part (i) to predict the drying time of the varnish when 6.5 gm of additive is being used.

OR

(b) A tobacco company claims that there is no relationship between smoking and lung ailments. To investigate the claims random sample of 300 males in the age group of 40 to 50 is given medical test. The observed sample results are tabulated below : (CO1, CO2)

	Lung ailment	No lung ailment	Total
Smokers	75	105	180
Non-Smokers	25	95	120
Total	100	200	300

Tabulated value at 5% level of significance and one degree of freedom is 3.841.

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TPE–401

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

PETROLEUM GEOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Describe Sedimentary Basins of India and also write all sedimentary basins of India (category wise). (CO1)

OR

(b) What is the mode of occurrences of Hydrocarbons and also explain surface occurrence of hydrocarbon. (CO1)

2. (a) (i) Write a short note on Inorganic Theory of Hydrocarbons.

(ii) Describe clastic and carbonate reservoir rocks.

(CO1, CO2, CO3)

OR

(b) Explain the concept of Migration of Hydrocarbon and also describe Primary and Secondary Migration. (CO1, CO2, CO3)

3. (a) Define Reservoir Traps. What are the different types of reservoir traps ? With the help of a neat sketch, explain stratigraphic traps. (CO1, CO3)

OR

(b) Define what is a source rock and why is it necessary. Describe types of source rock and the concept of potential, effective, relic effective and spent source rock. (CO1, CO3)

4. (a) Explain in brief, what different chemical constituents presents in crude oil. (CO2)

OR

(b) Describe all physical properties of crude oil. (CO2)

5. (a) Explain paraffinic and naphthenic class of crude oil. (CO1, CO2)

OR

(b) Describe carbonate and siliciclastic reservoir rock. (CO1, CO2)

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TPE–403

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

HEAT AND MASS TRANSFER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the modes of heat transfer ? Define Fourier's law of conduction. How conductivity of the material varies with temperature in solid and gas ? Derive an expression for the rate of heat transfer for the single plane wall. (CO1)

OR

(b) Illustrate with mathematical expression for the logarithmic mean area of the plane wall and cylindrical shape where the rate of heat transfer taking place.

Hence calculate the 'A_m' for the following :

A hollow cylindrical having inner and outer diameter as 10 and 20 cm, the length of the cylinder is 50 cm. (CO1)

P. T. O.

2. (a) What do you mean by insulation and critical radius of insulation ? Derive an expressions for the critical radius of insulation for the sphere. Specify the symbol used in the derivation. (CO2)

OR

(b) A gas filled tube has 4 mm inside diameter and 50 cm length. The gas is heated by an electrical wire of diameter 100 microns located along the axis of the tube. Current and voltage drop against the heating element are 1 amps and 8 volts, respectively. If the measured wire and inside tube wall temperature are 400°C and 350°C respectively. Determine the thermal conductivity of the gas filled in the tube. (CO2)

3. (a) An electric hot plate is maintained at a temperature of 350 °C, and is used to keep a solution boiling at 95 °C. The solution is contained in a cast iron vessel of wall thickness 25 mm, which is enameled inside to a thickness of 0.8 mm. The heat transfer coefficient for the boiling solution is 5.5 kW/m^{2}K, and the thermal conductivity of the cast iron and enamel are 50 and 1.05 W/m.K, respectively. Calculate the overall heat transfer coefficient and the rate of heat transfer per unit area. (CO2)

OR

(b) The wall of a cold storage consists of three layers–an outer layer of ordinary bricks, 25 cm thick, a middle layer of cork, 10 cm thick and an inner layer of cement, 6 cm thick. The thermal conductivities of materials are-brick : 0.7, cork : 0.043, and cement : 0.72 W/m°C. The temperature of the outer surface wall is 30°C and that of the inner is – 15°C. Calculate (i) the steady state rate of heat gain per unit area of the wall, (ii) the temperature at the interfaces of the composite wall, and (iii) the percentage of the total heat transfer resistance offered by the individual layers. What additional thickness of cork should be provided to make the rate of heat transfer 30 percent less than the present value ?

(CO2)

4. (a) What are the types of heat exchanger on the basis of nature of (i) heat exchange process, relative (ii) relative direction of fluid motion and (iii) Design and constructional features ? (CO1, CO2)

OR

(b) In a certain double pipe heat exchanger hot water flows at arate of 5000 kg/h and gets cooled from 95°C to 65°C. At the same time 50000 kg/h of cooling water at 30°C enters the heat exchanger. The flow conditions are such that overall heat transfer coefficient remains constant at 2270 W/m^{2}K. Determine the heat transfer area required and the effectiveness, assuming two streams are in parallel flow. Assume for both the streams $C_p = 4.2$ kJ/kg-K. (CO1, CO2)

5. (a) Derive the relation of logarithmic mean temperature difference [LMTD] for counter flow heat exchanger. Specify the symbol used. (CO2)

OR

(b) Illustrate the following : (CO2)

(i) LMTD

(ii) Effectiveness

(iii) Capacity ratio

(iv) NTU

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TPE–403

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

HEAT AND MASS TRANSFER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the modes of heat transfer ? Define Fourier's law of conduction. How conductivity of the material varies with temperature in solid and gas ? Derive an expression for the rate of heat transfer for the single plane wall. (CO1)

OR

(b) Illustrate with mathematical expression for the logarithmic mean area of the plane wall and cylindrical shape where the rate of heat transfer taking place.

Hence calculate the 'A_m' for the following :

A hollow cylindrical having inner and outer diameter as 10 and 20 cm, the length of the cylinder is 50 cm. (CO1)

P. T. O.

2. (a) What do you mean by insulation and critical radius of insulation ? Derive an expressions for the critical radius of insulation for the sphere. Specify the symbol used in the derivation. (CO2)

OR

(b) A gas filled tube has 4 mm inside diameter and 50 cm length. The gas is heated by an electrical wire of diameter 100 microns located along the axis of the tube. Current and voltage drop against the heating element are 1 amps and 8 volts, respectively. If the measured wire and inside tube wall temperature are 400°C and 350°C respectively. Determine the thermal conductivity of the gas filled in the tube. (CO2)

3. (a) An electric hot plate is maintained at a temperature of 350 °C, and is used to keep a solution boiling at 95 °C. The solution is contained in a cast iron vessel of wall thickness 25 mm, which is enameled inside to a thickness of 0.8 mm. The heat transfer coefficient for the boiling solution is 5.5 kW/m^{2}K, and the thermal conductivity of the cast iron and enamel are 50 and 1.05 W/m.K, respectively. Calculate the overall heat transfer coefficient and the rate of heat transfer per unit area. (CO2)

OR

(b) The wall of a cold storage consists of three layers–an outer layer of ordinary bricks, 25 cm thick, a middle layer of cork, 10 cm thick and an inner layer of cement, 6 cm thick. The thermal conductivities of materials are-brick : 0.7, cork : 0.043, and cement : 0.72 W/m°C. The temperature of the outer surface wall is 30°C and that of the inner is – 15°C. Calculate (i) the steady state rate of heat gain per unit area of the wall, (ii) the temperature at the interfaces of the composite wall, and (iii) the percentage of the total heat transfer resistance offered by the individual layers. What additional thickness of cork should be provided to make the rate of heat transfer 30 percent less than the present value ?

(CO2)

4. (a) What are the types of heat exchanger on the basis of nature of (i) heat exchange process, relative (ii) relative direction of fluid motion and (iii) Design and constructional features ? (CO1, CO2)

OR

(b) In a certain double pipe heat exchanger hot water flows at arate of 5000 kg/h and gets cooled from 95°C to 65°C. At the same time 50000 kg/h of cooling water at 30°C enters the heat exchanger. The flow conditions are such that overall heat transfer coefficient remains constant at 2270 W/m^2K. Determine the heat transfer area required and the effectiveness, assuming two streams are in parallel flow. Assume for both the streams $C_p = 4.2$ kJ/kg-K. (CO1, CO2)

5. (a) Derive the relation of logarithmic mean temperature difference [LMTD] for counter flow heat exchanger. Specify the symbol used. (CO2)

OR

(b) Illustrate the following : (CO2)

(i) LMTD

(ii) Effectiveness

(iii) Capacity ratio

(iv) NTU

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TPE–404

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

APPLIED SEDIMENTOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Water flowing through a pipe has a dynamic viscosity of 8.9×10^{-4} Pa.s and density of 1000 kg/m^3. The pipe has an internal diameter of .025 m and the water flows at an average speed of .15 m/s. What is the Reynolds' number for the flow ?

(ii) Explain Diagenesis. (CO1)

OR

(b) What are the differences between physical weathering and chemical weathering ? What are the main agents responsible for physical weathering and chemical weathering ? (CO1)

P. T. O.

2. (a) (i) A syrup flows at a rate of $4.0\ m^3s^{-1}$ in a circular pipe of 6 cm, syrup has a density of $1300\ kgm^{-3}$ and viscosity of 17 Pass. Find whether the flow is laminar or turbulent.

   (ii) Describe mode of sediment transport. (CO1, CO2)

OR

   (b) Explain Sediments packing, Permeability and types of permeability. (CO1, CO2)

3. (a) (i) Describe Reynolds' number and with the help of neat diagram, explain Laminar and Turbulent flow.

   (ii) If a fluid with a viscosity of $0.4\ Ns/m^2$ and relative density of $900\ kg/m^3$ flows through a pipe of 20 mm with a velocity of 2.5 m, calculate the Reynolds' number. (CO1, CO2)

OR

   (b) With the help of table explain, Wentworth grain size classification (CO1, CO2)

4. (a) Explain what clastic sedimentary rock is. Write the name of two clastic sedimentary rocks. Give the classification of clastic sedimentary rocks. (CO1, CO2)

OR

   (b) What is Physical Weathering ? Describe three processes that contribute to the weathering of rocks. (CO1, CO2)

5. (a) Explain what do you understand by kurtosis and what does it represents ? Draw the kurtosis curve for (i) sorted (ii) poor sorted (iii) well sorted. (CO1, CO2)

OR

   (b) Explain porosity and types of porosity. Discuss all the parameters on which porosity of rocks depend. (CO1, CO2)

Roll No. ....

TPE–402

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

RESERVOIR ENGINEERING—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) What are the different factors that affect porosity ? Explain.

(ii) Using the Archimedes method, calculate the pore and bulk volumes and the porosity :

Dry weight of sample = 489.3 g

Weight of sample saturated with water = 510.6 g

Density of water = 1.0 g/cm^3

Weight of saturated sample submerged in water = 269.6 g

(CO1)

P. T. O.

OR

(b) Calculate the arithmetic average and thickness-weighted average from the following measurements : (CO1)

Sample	Thickness, ft	Porosity, %
1	1.0	10
2	1.5	12
3	1.5	15
4	2.0	14
5	2.1	11
6	1.8	12

2. (a) (i) Describe what is a Reservoir and Reservoir Engineering. Differentiate between conventional and unconventional reservoir.

(ii) Write short notes on the following :

(1) Cricondentherm

(2) phase envelope

(3) bubble point curve

(4) Dew point curve

(CO1)

OR

(b) Explain all physical properties of crude oil. (CO1)

3. (a) With the help of phase diagram, differentiate between retrograde gas condensate and near critical gas condensate reservoir. (CO1)

OR

(b) A core sample coated with paraffin immersed in a container of liquid displaced 10.9 cm^3 of the liquid. The weight of the dry core sample was 20.0 g, while the weight of the dry sample coated with paraffin was 20.9 g. Assume the density of the solid paraffin is 0.9 g/cm^3.

Calculate the bulk volume of the sample. (CO1)

4. (a) With the help of a table, explain tissot classification of crude oils. (CO1, CO2)

OR

(b) (i) Write short notes on the following : (CO1, CO2)

(1) Gas cap reservoir

(2) Sorting

(ii) Explain cap rock and porosity and permeability and also differentiate between primary and secondary porosity.

5. (a) Explain paraffinic and naphthenic class of crude oil. (CO2)

OR

(b) Explain any *two* petro-physical properties of rocks. (CO2)

Roll No. ....

TPE–405

B. TECH. (PETROLEUM ENGINEERING)
(FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

PETROLEUM DRILLING ENGINEERING—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the objectives of well planning ? What parameters to be considered in well planning for a vertical well ? (CO1)

OR

(b) What are the main systems of a rotary drilling rig ? Describe the purpose and components of hosting system of a rotary drilling rig.(CO1)

2. (a) What is drilling line ? What are the considerations in drilling line design ? Discuss. (CO1)

OR

(b) A rig must hoist load of 3,00,000 lbf. The drawworks can provide an input power to the block and tackle system as high as 500 h.p. Eight

P. T. O.

lines are strung between the crown block and travelling block. Calculate :

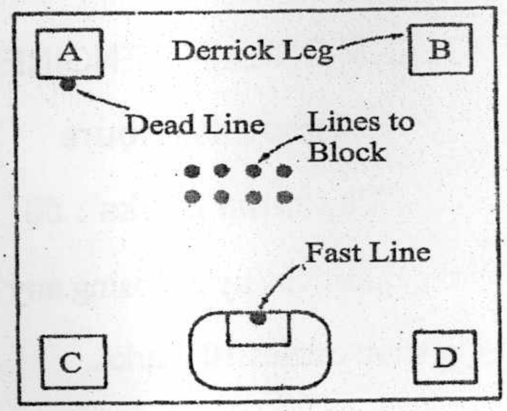
(i) the static tension in the fast line when upward motion is impending

(ii) the maximum hook horsepower available

(iii) the maximum hoisting speed

(iv) the actual derrick load

Assume that the rig floor is arranged as shown in Fig. below : (CO1)



Projections of drilling line on rig floor

3. (a) Describe the purpose and components of circulatory system of a rotary drilling rig. (CO1)

OR

(b) Calculate the liner size required for a double-acting duplex pump where rod diameter is 2.5 in, stroke length is 22 in stroke, pump speed is 70 strokes/min. In addition, the maximum available pump hydraulic horsepower is 1200 h.p. For optimum hydraulics, the pump recommended delivery pressure is 3,000 psi. Assume the volumetric efficiency of pump is 98%. (CO2)

4. (a) What are different types of drilling fluid ? What are the functions of drilling fluids ? (CO3)

OR

(b) During a drilling rig structure fatigue test, the operator measured the wind load of 0.5 psi. The rig has ten lines which are strung through the traveling block. A hook load of 2,50,000 lbf is being hoisted. According to the API standard, calculate the wind velocity, and the total compressive load. (CO3)

5. (a) What are the drilling fluid additives ? The emulsifier is used as an additive in drilling fluid. What role emulsifiers play in drilling fluid ? Name the *two* common emulsifiers. (CO3)

OR

(b) A 15-in bit is used to drill a hole at a rate of 80 ft/hr. where the porosity of the formation is 20%. Calculate the solid volume generated in this drilling operation. If the density of the solid is 910 lbm/bbl, calculate the solid generation in tons/hr. also. (CO3)